SME Flow — DApp Demonstration Script

00:00–01:05 | Role Registration and Approval

Hello everyone. In this video, we will demonstrate SME Flow, our blockchain-based accounts receivable financing platform for small and medium-sized enterprises.

Our application runs on the Ethereum Sepolia test network and uses MetaMask to sign blockchain transactions.

First, we demonstrate role-based access control.

The Supplier requests permission to use the platform. The request is recorded on the blockchain and remains pending until the Administrator approves it.

After approval, the Supplier role becomes active, indicated by the green status indicator.

This process is managed by our RoleRegistry smart contract, which ensures that only authorized participants can perform specific operations.

01:05–02:20 | Invoice Creation

Next, the Supplier creates a new invoice.

The invoice includes the Buyer's wallet address, invoice number, invoice amount, issue date, and payment due date.

In this demonstration, the invoice value is 0.1 ETH.

When the Supplier submits the invoice, the transaction is signed through MetaMask and recorded on the Sepolia blockchain.

Our InvoiceRegistry smart contract manages invoice information and prevents duplicate registrations within the system.

The invoice is initially waiting for Buyer confirmation.

02:20–03:05 | Buyer Confirmation

Now we switch to the Buyer account.

The Buyer can review the invoice details, including the Supplier, invoice amount, and payment deadline.

After checking the information, the Buyer confirms the invoice through MetaMask.

The invoice status changes to Confirmed.

This is an important security feature because the Supplier cannot request financing using an invoice that has not been confirmed by the designated Buyer.

03:05–03:40 | Financing Request

After confirmation, the Supplier can use the invoice to apply for financing.

Here, the Supplier requests 0.08 ETH against the 0.1 ETH invoice.

The maximum annual interest rate is set to 8%, and the holdback ratio is set to 10%.

Holdback means that part of the financing amount is temporarily retained by the smart contract as protection against potential losses.

The financing request is recorded on the blockchain and becomes available to potential Funders.

03:40–04:35 | Funder Submits an Offer

Next, we switch to the Funder account.

The Funder can view available financing opportunities and examine the financing amount and maximum interest rate.

In this case, the Funder submits an offer with a 6% annual interest rate.

Since this rate is below the Supplier's maximum rate of 8%, the offer satisfies the financing conditions.

The offer is submitted through the FinancingMarket smart contract and becomes visible to the Supplier.

04:35–05:05 | Supplier Accepts the Offer

Now we return to the Supplier account.

The Supplier reviews the available quotation and accepts the 6% offer.

After confirmation, the financing status changes to Offer Accepted.

At this stage, the financing terms have been agreed upon, but the actual funds have not yet been transferred.

Only the selected Funder is authorized to complete the funding transaction.

05:05–05:50 | Funder Provides Financing

The selected Funder now transfers 0.08 ETH to the FinancingPool smart contract.

The smart contract automatically calculates how the funds should be distributed.

With a 10% holdback, 0.008 ETH is retained in the contract, while 0.072 ETH is transferred immediately to the Supplier.

The Supplier therefore receives early liquidity without waiting for the Buyer to pay the invoice.

Once the transaction is confirmed, the financing status becomes Funded.

This demonstrates how our system automates financing and reduces the need for manual fund allocation.

05:50–06:45 | Dispute Creation and Evidence Submission

Next, we demonstrate our dispute resolution mechanism.

We switch to the Buyer account and use the mobile layout to demonstrate the responsive interface.

The Buyer opens a dispute related to the funded invoice and submits supporting evidence.

Instead of storing the original evidence directly on the blockchain, the system stores a cryptographic hash.

This provides a record that can later be used to check whether the evidence has been modified, without exposing the original content on-chain.

Once the dispute is opened, the related financing operations are frozen until the dispute is resolved.

06:45–07:20 | Repayment During a Dispute

Although the invoice is under dispute, the Buyer can still make the required repayment.

Here, the Buyer transfers the full invoice amount of 0.1 ETH to the FinancingPool contract.

However, because the dispute remains open, the contract does not immediately distribute the repayment.

Instead, the payment is held in escrow.

The invoice status changes to Repayment Deposited.

This mechanism protects the funds while allowing the Buyer to complete the payment obligation.

07:20–08:15 | Arbitration and Automatic Settlement

We now switch to the Arbitrator account.

The Arbitrator can review the dispute status, the submitted evidence hashes, and the relevant blockchain records.

In this demonstration, the Arbitrator selects Resume Workflow.

The smart contract then resolves the dispute, releases the frozen repayment, and automatically distributes the funds according to the previously agreed financing terms.

The Funder receives the principal and interest, the platform receives its fee, and the remaining funds are transferred to the Supplier.

The invoice status changes to Repaid, and the financing status becomes Settled.

This demonstrates how smart contracts enforce predefined financial rules without allowing the Arbitrator to choose arbitrary payment amounts.

08:15–09:28 | Auditor and Transaction History

Finally, we switch to the Auditor role and return to the desktop interface.

The Auditor can review the complete financing process without modifying business records.

The dashboard provides information about financed invoices, funding amounts, Buyer repayments, platform fees, and financial allocations.

We can also view the transaction history and blockchain events associated with the invoice.

Each blockchain transaction can be independently verified through Etherscan.

This provides transparency and traceability for Suppliers, Buyers, Funders, and Auditors.

Conclusion

In this demonstration, we have completed an end-to-end accounts receivable financing process, including role approval, invoice confirmation, financing, funding, dispute resolution, repayment, automatic settlement, and auditing.

By combining smart contracts, MetaMask, a web interface, and backend services, SME Flow provides a transparent and structured approach to SME receivables financing.

Thank you for watching.

SME Flow — 区块链应收账款融资平台演示讲解稿

00:00–01:05｜角色申请与权限审批

大家好。在本次视频中，我们将演示 SME Flow，一个面向中小企业的区块链应收账款融资平台。

我们的应用运行在以太坊 Sepolia 测试网络上，并使用 MetaMask 钱包对区块链交易进行签名。

首先，我们展示系统的角色权限管理功能。

供应商首先向平台申请 Supplier 角色权限。申请会被记录在区块链上，在管理员批准之前，角色状态将保持为待审核。

管理员批准申请后，Supplier 角色被激活，界面上的状态标识变为绿色。

这一过程由 RoleRegistry 智能合约管理，确保只有经过授权的参与者才能执行相应的业务操作。

01:05–02:20｜创建发票

接下来，供应商创建一张新的发票。

发票信息包括买方的钱包地址、发票编号、发票金额、开具日期以及付款截止日期。

在本次演示中，发票的面值为 0.1 ETH。

供应商提交发票时，需要通过 MetaMask 签署交易，并将相关信息记录到 Sepolia 区块链上。

我们的 InvoiceRegistry 智能合约负责管理发票信息，并防止同一张发票在系统内部被重复登记。

此时，发票状态为等待买方确认。

02:20–03:05｜买方确认发票

现在，我们切换到 Buyer，也就是买方账户。

买方可以查看发票的详细信息，包括供应商、发票金额以及付款截止日期。

在核对信息后，买方通过 MetaMask 确认这张发票。

交易完成后，发票状态变为 Confirmed，也就是已确认。

这是系统一项重要的安全机制，因为只有经过指定买方确认的发票，才能被供应商用于申请融资。

03:05–03:40｜发起融资申请

发票确认后，供应商便可以使用这笔应收账款申请融资。

在这里，供应商使用一张面值为 0.1 ETH 的发票，申请获得 0.08 ETH 的融资。

最高年化利率设置为 8%，同时设置 10% 的 Holdback，也就是风险留存比例。

Holdback 的含义是，智能合约会暂时保留一部分融资资金，用于应对潜在的损失风险。

融资申请提交后，会被记录在区块链上，并向符合条件的资金提供方开放。

03:40–04:35｜资金提供方提交报价

接下来，我们切换到 Funder，也就是资金提供方账户。

资金提供方可以浏览当前开放的融资机会，并查看融资金额和最高利率等信息。

在本次演示中，Funder 提交了一个年化利率为 6% 的融资报价。

由于这一利率低于供应商设定的 8% 上限，因此符合融资条件。

报价通过 FinancingMarket 智能合约提交，并显示给供应商进行选择。

04:35–05:05｜供应商接受融资报价

现在，我们重新切换到 Supplier 账户。

供应商查看 Funder 提交的报价，并选择接受年化利率为 6% 的方案。

交易确认后，融资状态变为 Offer Accepted，也就是报价已接受。

需要注意的是，此时双方只是确定了融资条件，资金还没有真正转移。

只有被选中的 Funder 才有权限完成后续的放款交易。

05:05–05:50｜资金提供方完成放款

被选中的 Funder 现在向 FinancingPool 智能合约转入 0.08 ETH。

智能合约会根据预先设定的规则，自动计算并分配这笔资金。

由于 Holdback 比例为 10%，其中 0.008 ETH 会暂时保留在智能合约中，而另外 0.072 ETH 将立即支付给供应商。

这样，供应商就可以在买方尚未支付发票款项之前，提前获得所需的流动资金。

交易确认后，融资状态变为 Funded，也就是已放款。

这一过程展示了智能合约如何自动执行融资资金的分配，从而减少人工操作。

05:50–06:45｜发起争议与提交证据

接下来，我们展示平台的争议解决机制。

我们切换到 Buyer 账户，并使用移动端布局，展示系统的响应式界面。

买方针对已经完成融资的发票发起争议，并提交相关证据。

系统不会直接将原始证据内容保存在区块链上，而是保存证据的加密哈希值。

这样既能够为后续验证证据是否被修改提供依据，也避免了将原始内容直接公开到链上。

争议发起后，相关融资结算操作会暂时被冻结，直到争议得到解决。

06:45–07:20｜争议期间的还款

虽然发票目前处于争议状态，但买方仍然可以履行付款义务。

在这里，买方向 FinancingPool 智能合约支付完整的发票金额，也就是 0.1 ETH。

但是，由于争议尚未解决，智能合约不会立即将这笔款项分配给供应商、资金提供方和平台。

相反，资金会暂时保留在合约中，进入托管状态。

此时，发票状态变为 Repayment Deposited，也就是还款已存入托管。

这一机制能够在允许买方履行付款义务的同时，避免争议解决前发生不可逆的资金分配。

07:20–08:15｜仲裁与自动结算

现在，我们切换到 Arbitrator，也就是仲裁员账户。

仲裁员可以查看争议状态、提交的证据哈希，以及相关的区块链记录。

在本次演示中，仲裁员选择 Resume Workflow，也就是恢复业务流程。

随后，智能合约会解除争议冻结，并根据之前已经确定的融资条件，自动分配托管资金。

资金提供方将收到融资本金和相应利息，平台获得服务费用，剩余资金则支付给供应商。

交易完成后，发票状态变为 Repaid，也就是已还款；融资状态变为 Settled，也就是已结算。

这一过程展示了智能合约如何根据预先定义的金融规则完成结算，而不是由仲裁员任意决定付款金额。

08:15–09:28｜审计与交易记录查询

最后，我们切换到 Auditor，也就是审计员角色，并恢复桌面端界面。

审计员可以查看完整的融资流程，但不能修改系统中的业务记录。

平台的审计仪表盘展示了融资发票、放款金额、买方还款、平台费用以及资金分配等信息。

我们还可以查看与该发票相关的交易历史和区块链事件。

每一笔区块链交易都可以通过 Etherscan 进行独立核实。

这为供应商、买方、资金提供方和审计员提供了透明、可追溯的业务记录。

结束总结

通过本次演示，我们完整展示了一笔应收账款融资业务的全过程，包括角色审批、发票确认、融资申请、放款、争议解决、还款、自动结算以及审计。

通过结合智能合约、MetaMask、Web 前端和后端服务，SME Flow 为中小企业应收账款融资提供了一种透明、结构化的解决方案。

感谢观看！